Neerav Chaudhary

APPLIED AI / ML ENGINEER | INDUSTRIAL ML, LLM EVALUATION, HEALTHCARE AI

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LinkedIn | GitHub | Portfolio

Applied AI/ML engineer and published researcher delivering measurable results in industrial compliance, employee operations, and healthcare AI. Built systems that restored regulatory data uptime to 100%, cut compliance reporting from nine hours to 22 seconds, completed more than 92% of held-out agent workflows, and improved emergency-guidance coverage by 20.5%.

PROFESSIONAL EXPERIENCE

NTPC | AI/ML Engineer Intern **Nov 2023-May 2024**

- Reduced regulatory and wrongful-citation risk with an XGBoost + SHAP model that separated plant-stack PM2.5 spikes from traffic and regional background; achieved R2 = 0.84 with source attribution validated at r = 0.93-0.96 against ground truth.
- Enabled pre-emptive curtailment with 24-hour exceedance forecasts at RMSE 9.9 ug/m3, while a Random Forest virtual sensor filled maintenance gaps at MAPE 4.8% and lifted regulatory data uptime from 92.16% to 100%.
- Cut incident-to-report turnaround from approximately nine hours to 22 seconds by converting model attributions into decision-ready internal briefs and regulator letters with locally deployed Llama 3.1.
- De-risked procurement for INR 1-2 crore-per-unit CAAQMS stations by matching pollutant coverage, 1-15 minute sampling, plant needs, and CPCB requirements against lower-cost IoT alternatives.

Deep Strat | AI Safety and Research Intern **Aug-Nov 2023**

- Compared Random Forest and CNN models on 2,000 customer records against a cost-sensitive objective; reduced false churn alerts by 25% while maintaining AUC and focused retention effort on higher-risk customers.
- Turned unstructured complaints into sentiment- and intent-classified, correctly routed, SOP-grounded cases through a three-stage LLM workflow, giving operations teams consistent cases they could review and act on.
- Advised founders and policy researchers on model failure modes, guardrails, and responsible AI; authored a 50-page technical reference that supported jointly authored AI-safety research presented at Nullcon.
- Created the firm's internal AI curriculum from classical ML through embeddings, attention, hallucination, bias, and RLHF, giving analysts and leadership a shared framework for evaluating AI systems.

Manipal University Jaipur | Medical Imaging Researcher **Aug-Dec 2022**

- Co-authored an IEEE ICRITO paper and built its Python/OpenCV pipeline across 2,198 ADNI MRI images; focusing measurement on the hippocampal region improved reported accuracy from 66% to 71%.
- Built a separate four-class TensorFlow experiment on 6,400 MRI images; probability-level ensembling improved held-out accuracy from 78% to 85%.

SELECTED AI PROJECTS

Relay - Voice-first employee operations | React, TypeScript, Fastify, Deepgram

- Removed portal-hunting across People Ops, IT, Finance, and Administration by unifying seven employee workflow families in one authenticated voice-and-text workspace.
- Across 252 conversations and 764 turns, completed 103/110 development workflows (93.6%) and 39/42 held-out rewordings (92.9%).
- Reached the correct employee record in 186/187 scored tool runs (99.5%), blocked all 30 fabricated record IDs, and made zero unsafe state changes across 21 adversarial runs.

TraceLens - Healthcare AI safety evaluation | Python, PyTorch, Hugging Face

- Audited emergency-response quality across 277 healthcare cases, exposing 31 emergencies missed by the baseline chatbot and testing false alarms against 134 non-emergency cases.
- Built a held-out emergency detector with AUC 0.966 that caught and disclosed 18 of 31 emergencies missed by the original chatbot - a 58% recovery rate.
- Raised correct final emergency guidance from 39/70 to 47/70 (+8 cases; +20.5% relative) while adding one false alarm across 134 non-emergency cases.

PUBLICATION AND RESEARCH

IEEE ICRITO 2022 | First Author, Computer Vision Research

- Published an interpretable atrophy measure that converted visible tissue and dark-space ratios into cohort reference bounds of 0.41 and 0.83, making each classification traceable to image-level evidence.

N. Chaudhary and S. S. Verma. "Analyzing Brain Atrophy in Dementia Using Computer Vision." doi:10.1109/ICRITO56286.2022.9964994

EDUCATION

San Jose State University | M.S., Industrial and Engineering Management | GPA 3.81 **2024-2026**
Manipal University Jaipur | B.Tech., Computer Engineering | GPA 8.8/10 **2018-2022**

SKILLS

AI and LLMs: LLM evaluation, voice agents, RAG, structured tool calling, guardrails, Hugging Face, Deepgram, LangChain, llama.cpp, Ollama
Engineering: Python, TypeScript, React, FastAPI, Fastify, SQL, PostgreSQL, SQLite, REST APIs, Docker, AWS, Git
ML and research: XGBoost, SHAP, LSTM, Random Forest, PyTorch, TensorFlow, scikit-learn, OpenCV, forecasting, attribution, calibration